

The empirical recurrence for OEIS A251494: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A251494 records “Empirical recurrence of order 52”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 52, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 52 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A251494 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with every 2×2 subblock summing to 4 5 6 7 or 8.”. It has offset 1 and begins

27844, 3361296, 411510096, 50511532876, 6203258853660, 761894242946484, 93579182156013156, 1149386158385

The entry states:

Empirical recurrence of order 52 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:25:47, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 52 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{52} c_i a(n-i),$$

c_1	161	c_{14}	1253332953080641	c_{27}	−229430903046705402626	c_{40}	46936331122655
c_2	−4146	c_{15}	26095778872493510	c_{28}	−332055505305503208018	c_{41}	23886147749633
c_3	−99086	c_{16}	−48654625714125195	c_{29}	440325723529501943704	c_{42}	−8199928112280
c_4	4025719	c_{17}	−260176131205401648	c_{30}	281414846009547740281	c_{43}	−2072109081225
c_5	3352718	c_{18}	730743832771969293	c_{31}	−525294895032204161992	c_{44}	8500608837212
c_6	−1143269351	c_{19}	1502063360489568965	c_{32}	−127685608591143631767	c_{45}	961368165113
c_7	6323065419	c_{20}	−6225510155500329353	c_{33}	408889884173453304811	c_{46}	−503142220977
c_8	111830788127	c_{21}	−4116086371165437899	c_{34}	7499369422803642169	c_{47}	−17484399941
c_9	−946832349806	c_{22}	32877090960710570170	c_{35}	−211643864068737289778	c_{48}	156119569661
c_{10}	−4245862352721	c_{23}	−3198116216306014279	c_{36}	25656578512266508379	c_{49}	−6772937195
c_{11}	53521620702661	c_{24}	−110881563115778371997	c_{37}	72974520805344924476	c_{50}	−2193692295
c_{12}	46473371138334	c_{25}	64601914740248236032	c_{38}	−15672822621732300540	c_{51}	3125595340
c_{13}	−1558596393101887	c_{26}	240410371748256608994	c_{39}	−16539948067063790336	c_{52}	1019215872

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 52, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $52 + 4$ terms removed returns order 52 as well, so $\deg q_{\min} = 52 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $52 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 52 that this sequence satisfies — and that family turns out to have exactly one member.

References

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